

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

Throughout EEL 3701, we will be using Intel’s Quartus Prime software suite to design, simulate, and compile digital circuit designs for use on the DE10-Lite FPGA Development Board. The installation of Quartus Prime will take multiple steps. Make sure to install both Quartus and one simulator, either ModelSim or Questa. I recommend using ModelSim, as it has a far simpler setup and does not require a license. The Questa simulator has a more complicated setup and is even more complicated to use, but works better on Linux.

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Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

QUARTUS INSTALLATION INSTRUCTIONS

QUARTUS INSTALLATION FOR WINDOWS 10/11

- 1) Go to the [Quartus Prime Lite 25.1 Windows download page](#) and select the “Individual Files” tab of the website, as shown in the red rectangle. (Note that newer version may be available but have **NOT** been tested.)
- 2) Download the following (You may need to accept license agreements):
 - a. Quartus Prime
 - b. Intel MAX 10 FPGA Device Support
 - c. Intel MAX II, Intel MAX V Device Support
 - d. Intel Cyclone V Device Support

Individual Files

Quartus® Prime	Download QuartusLiteSetup-25.1std.0.1129-windows.exe	Size: 1.6 GB SHA1: 09e2c614d37d1553ae8c1e58e280b950460855da
Quarta®-Altera FPGA and Starter Editions	Download QuartaSetup-25.1std.0.1129-windows.exe	Size: 942.2 MB SHA1: 164348081f610cd652280d0e3f8a27d173b0cddf

Add-On and Stand-Alone Software

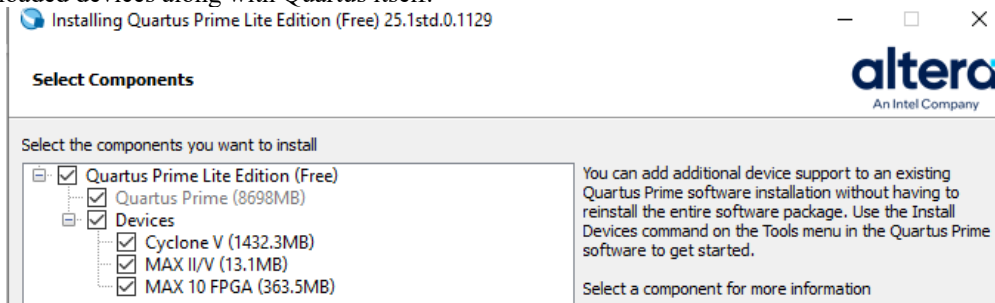
Ashling RiscFree IDE for Altera	Download RiscFreeSetup-25.1std.0.1129-windows.exe	Size: 869 MB SHA1: 8da1e9e29354ff53fb3de86c203c88ccdd8cd660
Intel® Quartus® Prime Driver Installer	Download QuartusDriversSetup-25.1std.0.1129-windows.exe	Size: 33.4 MB SHA1: 1b6c0c6aad8b21d7e499c7ce659f7468142c5d9c
Intel® Quartus® Prime Help	Download QuartusHelpSetup-25.1std.0.1129-windows.exe	Size: 278.8 MB SHA1: 7fad1301e810e7a7ae5b8ebd12376b9c9575a440
Intel® Quartus® Prime Programmer and Tools	Download QuartusProgrammerSetup-25.1std.0.1129-windows.exe	Size: 378.9 MB SHA1: 09c3987b197ee4d5c34434c704bbe853d516b190

Devices

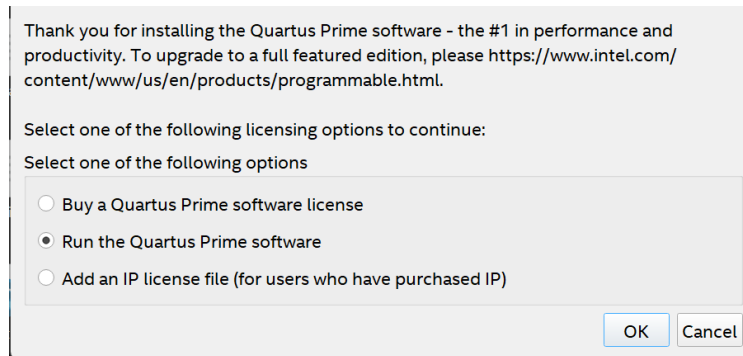
Arria® II device support	Download arria_lite-25.1std.0.1129.qdz	Size: 499.1 MB SHA1: d803f8a865d260f47cdb1f856c123fab5a9934bf
** Installation size: 0.52 GB		
Cyclone® IV device support	Download cyclone-25.1std.0.1129.qdz	Size: 466 MB SHA1: 835d2b1732549294eed625b692d044135499b5e8
** Installation size: 0.50 GB		
Cyclone® 10 LP device support	Download cyclone10lp-25.1std.0.1129.qdz	Size: 265.7 MB SHA1: b3cc2ca1b2e1b225407968e2f380e596df5b80d
** Installation size: 0.29 GB		
Cyclone® V device support	Download cyclonev-25.1std.0.1129.qdz	Size: 1.3 GB SHA1: a7225ec1bd36ccfd6826ea6273df5d21dd95633b
** Installation size: 1.40 GB		
MAX® II, MAX® V device support	Download max-25.1std.0.1129.qdz	Size: 11.4 MB SHA1: a3bc065b42a9d005f8d27fe45525411b49c13c43
** Installation size: 0.01 GB		
MAX® 10 FPGA device support	Download max10-25.1std.0.1129.qdz	Size: 288.1 MB SHA1: 8936dac1b092853f3b49a67fef22968ec10f8087
** Installation size: 0.35 GB		

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

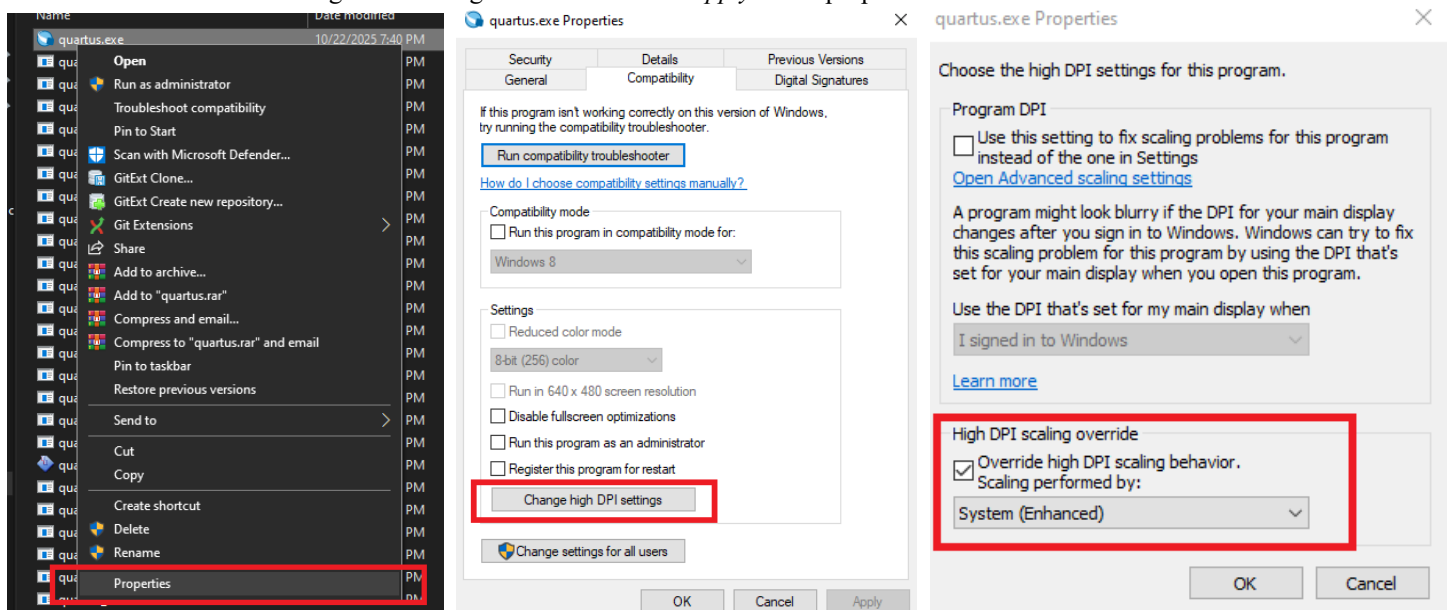
- 3) After the Quartus software has finished downloading, open the QuartusLiteSetup-25.1std.windows.exe file from the download to install the Quartus software. I suggest that you use all the defaults. After a few steps, you will see the image below that will install each of the downloaded devices along with Quartus itself.



- Windows may tell you that it is not safe to run this program with something like *Windows Protected Your PC*. Ignore this message. Select *More Info* and then select *Run Anyway*.
- Leave all default options in the installer and select Next as necessary.



- 4) When the installation is done, run the Quartus software.
- 5) If you have a High DPI display, some Quartus programs such as the simulation editor will not look correct. To fix this problem, navigate to C:\altera_lite\25.1std\quartus\bin64 and find quartus.exe. Right click on quartus.exe and choose *Properties*. Click on the *Compatibility* tab and click on the button labelled *Change high DPI settings*. In the window that appears, check the checkbox next to *Override high DPI scaling behavior* and choose the *System (Enhanced)* option in the dropdown menu. Click *OK* in the High DPI settings window and then *Apply* in the properties window.



- 6) Once you get your DE10-Lite, attach it to your computer and complete the driver installation as specified below.

Quartus Installation Instructions

(For Quartus v25.1 and ModelSim 19.1)

- a. Connect the USB-Blaster (DE10-Lite) to your PC.
- b. Open Device Manager.
- c. Locate USB-Blaster under Other devices.
- d. Right-click on USB-Blaster and select Update Driver.
- e. Choose Browse my computer for drivers.
- f. Select the Browse . . . button in the new window.
- g. Navigate to your Quartus installation directory:
- h. Default: C:\altera_lite\25.1std\quartus\drivers (Adjust to your Quartus version)
- i. If you can't find this directory, try clicking "*This PC*," then you should be able to find the C: drive or the altera lite folder.
- j. Note: Stop at the drivers folder, do not go deeper by opening a folder within the drivers folder.
- k. Confirm the file path and click Next.
- l. If prompted by Windows Security:
- m. Check the box for Always trust software from "Altera Corporation".
- n. Click Install.

When completed with the above, proceed to the **ModelSim Installation for Windows 10/11** section of this document (on page 8). ModelSim 19.1 is strongly recommended over Questa for students in 3701. Questa installation information is in the Appendices.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

QUARTUS INSTALLATION FOR LINUX

The installation directions provided in this document were designed for default Ubuntu 22.04 LTS, as this is the most recent distribution that is officially supported by Quartus.

- 1) Go to the [Quartus Prime Lite 25.1 Linux download page](#) and select the “Individual Files” tab of the website, as shown in the red rectangle. (Note that newer version may be available but should **NOT** be installed.)
- 2) Download the following (You may need to accept license agreements):
 - a. Intel Quartus Prime
 - b. Intel MAX 10 FPGA Device Support
 - c. Intel MAX II, Intel MAX V Device Support
 - d. Intel Cyclone V Device Support

Individual Files

Quartus® Prime	Download QuartusLiteSetup-25.1std.0.1129-linux.run	Size: 1.8 GB SHA1: ce0773469eacab5b7035c175484625f4ec3737d1
Questa®-Altera FPGA and Starter Editions	Download QuestaSetup-25.1std.0.1129-linux.run	Size: 3.1 GB SHA1: 149fe1e1cf253f2929804582c6cb658bca941dd5

Add-On and Stand-Alone Software

Ashling RiscFree IDE for Altera	Download RiscFreeSetup-25.1std.0.1129-linux.run	Size: 865.3 MB SHA1: 2d457bd18bfbf32f8f4037266b6c04853caed8e8
Intel® Quartus® Prime Driver Installer	Download QuartusDriversSetup-25.1std.0.1129-linux.run	Size: 9.1 MB SHA1: eb15977b79df7ddeb102e2ec182148068d273714
Intel® Quartus® Prime Help	Download QuartusHelpSetup-25.1std.0.1129-linux.run	Size: 276.5 MB SHA1: 078ec6422fe6bbdceecf1a7e683965a45e3e08411
Intel® Quartus® Prime Programmer and Tools	Download QuartusProgrammerSetup-25.1std.0.1129-linux.run	Size: 421.7 MB SHA1: b4e2d0ddaf969dbe01a255171327817682be019

Devices

Arria® II device support	Download arria_lite-25.1std.0.1129.qdz	Size: 499.1 MB SHA1: d803f8a865d260f47cdb1f856c123fab5a9934bf
** Installation size: 0.52 GB		
Cyclone® IV device support	Download cyclone-25.1std.0.1129.qdz	Size: 466 MB SHA1: 835d2b1732549294eed625b692d044135499b5e8
** Installation size: 0.50 GB		
Cyclone® 10 LP device support	Download cyclone10lp-25.1std.0.1129.qdz	Size: 265.7 MB SHA1: b3cc2ca1b2e1b225407968e2f380e596dff5b80d
** Installation size: 0.29 GB		
Cyclone® V device support	Download cyclonev-25.1std.0.1129.qdz	Size: 1.3 GB SHA1: a7225ec1bd36ccfd6826ea6273df5d21dd95633b
** Installation size: 1.40 GB		
MAX® II, MAX® V device support	Download max-25.1std.0.1129.qdz	Size: 11.4 MB SHA1: a3bc065b42a9d005f8d27fe45525411b49c13c43
** Installation size: 0.01 GB		
MAX® 10 FPGA device support	Download max10-25.1std.0.1129.qdz	Size: 288.1 MB SHA1: 8936dac1b092853f3b49a67fef22968ec10f8087
** Installation size: 0.36 GB		

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 3) Open a terminal and navigate to your Downloads folder. We can then run the Quartus installer by issuing the following commands. Make sure that you do not run the Quartus installer as root!

```
chmod +x QuartusLiteSetup-25.1std.0.1129-linux.run  
./QuartusLiteSetup-25.1std.0.1129-linux.run
```

Leave all default options in the installer and select Next each time it is necessary.

- 4) To allow easy access to your Quartus installation, create a text file called `quartus.desktop` using the text editor of your choice and copy the following contents into it. Make sure to change `YOURUSERNAME` to the username of the account that you installed Quartus for.

```
[Desktop Entry]  
Version=1.0  
Type=Application  
Terminal=false  
Exec=/home/YOURUSERNAME/altera_lite/25.1std/quartus/bin/quartus  
Name=Quartus  
Icon=/home/YOURUSERNAME/altera_lite/25.1std/quartus/adm/quartusii.png
```

- If your distribution supports desktop icons, you can copy this file to your Desktop folder and it will work as a desktop shortcut
 - To add a Quartus shortcut to your Apps menu, copy this file to `~/.local/share/applications`, creating that folder if necessary. This was tested for GNOME 44.
- 5) To allow Quartus to program your DE10-Lite board, we must add a udev rule enabling access to the USB-Blaster programming hardware on the DE10-Lite. Create the text file `/etc/udev/rules.d/51-altera-usb-blaster.rules` (requires superuser privileges) and populate it with the following:

```
SUBSYSTEM=="usb", ATTR{idVendor}=="09fb", ATTR{idProduct}=="6001", MODE="0666"  
SUBSYSTEM=="usb", ATTR{idVendor}=="09fb", ATTR{idProduct}=="6002", MODE="0666"  
SUBSYSTEM=="usb", ATTR{idVendor}=="09fb", ATTR{idProduct}=="6003", MODE="0666"  
SUBSYSTEM=="usb", ATTR{idVendor}=="09fb", ATTR{idProduct}=="6010", MODE="0666"  
SUBSYSTEM=="usb", ATTR{idVendor}=="09fb", ATTR{idProduct}=="6810", MODE="0666"
```

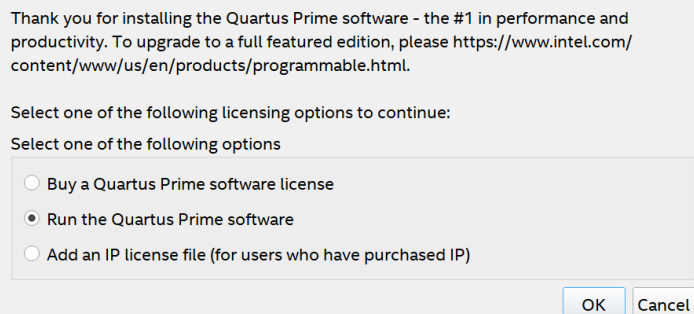
- 6) Restart your computer so that the new udev rules take effect.
- 7) Many Linux systems still encounter a permissions error when trying to connect to the USB Blaster, even after adjusting the udev rules. This shows up in an “Unable to read device chain – JTAG chain broken” error while using the programmer. This can be remediated by adjusting the JTAG daemon’s settings. Run the following in a terminal:

```
killall jtagd  
cd ~/altera_lite/25.1std/quartus/bin  
sudo mkdir /etc/jtagd  
sudo cp ../linux64/pgm_parts.txt /etc/jtagd/jtagd.pgm_parts  
./jtagd
```

- 8) You can verify that you are able to program your DE10-Lite by disconnecting and reconnecting it to your computer and running `./jtagconfig` in the same terminal you ran the above commands. If your setup is correct, you should see output that looks like:

```
1) USB-Blaster [1-2]  
031050DD 10M50DA (.|ES)/10M50DC
```

- 9) Run Quartus using either the `.desktop` file we made earlier or by running `~/altera_lite/25.1std/quartus/bin/quartus`. In the window that appears, select *Run the Quartus Prime Software* and then select *OK*.



- 10) Once you get your DE10-Lite, attach it to your computer and complete the driver installation as specified in https://www.terasic.com.tw/wiki/Altera_USB_Blaster_Driver_Installation_Instructions. When completed with the above, proceed to the [ModelSim Installation for Linux](#) instructions.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

QUARTUS INSTALLATION FOR INTEL MACS

Macs are **NOT** officially supported in this course (or by Altera!), so you may need to do additional troubleshooting to get everything working correctly. See the Windows tutorials for installing Quartus and a simulator for more information.

The best way to run Quartus on an Intel Mac is to dual boot Windows on your device by setting up Bootcamp. Students have also had success running the Windows version of Quartus through Parallels and other virtual machine software.

QUARTUS INSTALLATION FOR ARM MACS

Quartus is x86 software and it has no native Mac port, so the best way to get it running is to run it in a Linux VM with the FEX x86 emulator. I have chosen to use UTM (free), but you could also do this in Parallels (paid).

- 11) Download UTM from <https://docs.getutm.app/installation/macOS/>. You can download it for free by clicking Download from Github. Install UTM like you would install any other Mac application.
- 12) Ubuntu does not provide a generic Desktop ISO for ARM systems. download the Ubuntu 22.04 LTS Server (ARM64) installer. file should be called ubuntu-22.04.5-live-server-arm64.iso.
 - a. Open UTM.
 - b. Choose Create -> Virtualize -> Linux.
 - c. Select the downloaded ubuntu-22.04.5-live-server-arm64.iso.
 - d. Ensure the QEMU backend is used (not Apple Virtualization).
 - e. RAM: 4096+ MB. CPU: 1 core (multiple cores reduces emulated Quartus performance). Storage: 48+ GB
- 13) Start the virtual machine Start the virtual machine and install Ubuntu Server normally.
 - a. Do a minimal installation to ensure sufficient disk space for Quartus to install.
 - b. The installation will take 10-30 minutes depending on your computer.
- 14) After installation completes and you log in, install the Ubuntu desktop:

```
sudo apt update  
sudo apt install ubuntu-desktop
```

- 15) Once complete, reboot the VM to enter the graphical desktop. From the UTM VM info page, remove ubuntu-22.04.5-live-server-arm64.iso from the disk drive. Power on the VM and log in using the username and password you set up during the installation.
- 16) From the applications grid in the bottom left of the screen, open a terminal and run the following commands to update your system and install necessary packages. These commands will have you enter your password, and what you type for your password will be completely hidden. This is normal.

```
sudo apt update  
sudo apt upgrade  
sudo apt install curl
```

- 17) Install the FEX x86 emulator by running the following command (all one line). You will be prompted to enter your password.

```
curl --silent https://raw.githubusercontent.com/FEX-Emu/FEX/main/Scripts/InstallFEX.py \  
--output /tmp/InstallFEX.py && python3 /tmp/InstallFEX.py && rm /tmp/InstallFEX.py
```

- 18) The installer will prompt you for some options. You should have the installer download and install a rootfs automatically and set it as the default. You should be able to answer 'y' for every question it asks.
- 19) In a terminal, enter the FEX environment by typing FEXBash. You can now follow the Linux installation instructions to install Quartus within the FEXBash environment.
- 20) The only modification you need to make to the installation process is to insert FEXBash before the Exec command in the Quartus desktop entry if you decide to make one.

Anytime you want to run Quartus, you must start it from within a FEXBash environment.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

SIMULATOR INSTALLATION INSTRUCTIONS

MODELSIM INSTALLATION FOR WINDOWS 10/11

ModelSim is a free simulator that doesn't require a license to use. We will install ModelSim 19.1, as it doesn't throw errors for the deprecated parameters that Quartus inserts into waveforms.

- 1) Go to the [ModelSim-FPGAs Standard Edition Software Version 19.1](#) and select the "Windows Software" tab of the website. (Newer versions throw errors for the `-novopt` parameter and should **NOT** be installed).
- 2) You only need ModelSim-Intel FPGA Edition (includes Starter Edition).

ModelSim-FPGAs Standard Edition Software Version 19.1

 [Licensing Help](#)

Date- 03/31/2019

ModelSim-FPGA Standard Edition, Version 19.1std includes functional and security updates. Users should keep their software up-to-date and follow the [technical recommendations](#) to help improve security. Additional security updates are planned and will be provided as they become available. Users should promptly install the latest version upon release.

ModelSim-FPGA Standard Edition, Version 19.1std is subject to removal from the web when support for all devices in this release are available in a newer version, or all devices supported by this version are obsolete. If you would like to receive customer notifications by e-mail, please subscribe to our [subscribe to our customer notification mailing list](#).

Linux Software

Windows Software

Expand All

 [Download and Install Instructions](#)

ModelSim Software



ModelSim-FPGA Edition (includes Starter Edition)
(ModelSimSetup-19.1.0.670-windows.exe)

Size: 968.2 MB

Version: 19.1

 [Download](#)

OS: Microsoft Windows*

ECCN: EAR99

sha1: e3ebbf01c653892decdf7f18c2a3862ccc36954

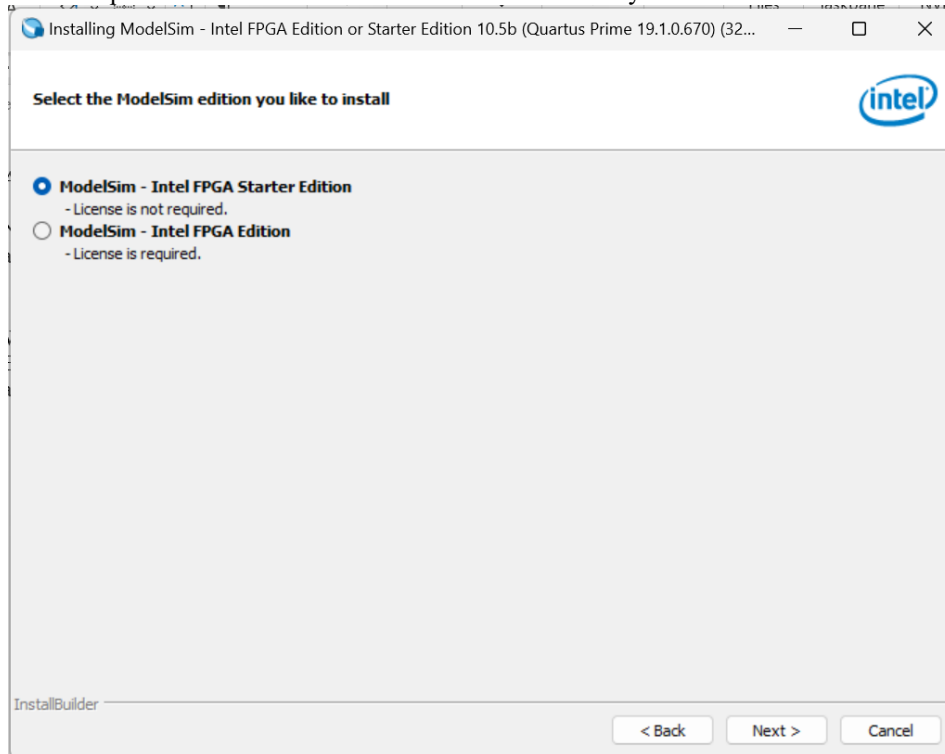
Download and Install Instructions:

1. Download the ModelSim-FPGA software into a temporary directory.
 2. Run the .exe file.
- Refer to the [Software Resources](#) page for more information, such as Community Support and Ecosystem.

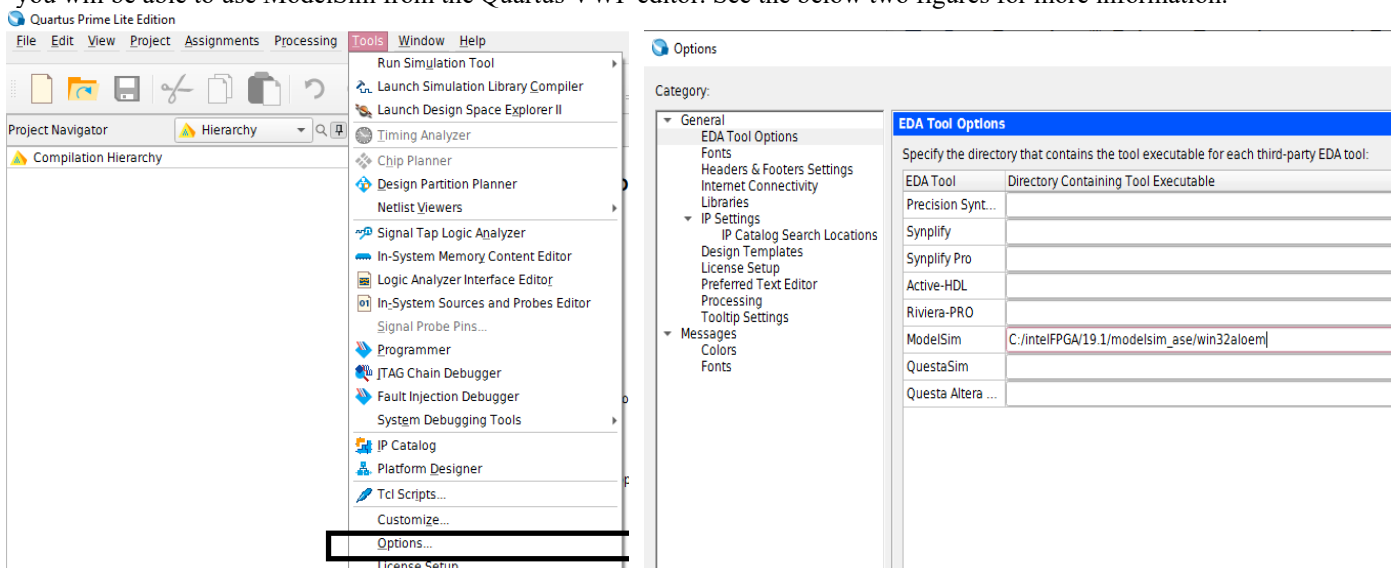
- 3) After ModelSim has finished downloading, run the following file to install it: ModelSimSetup-19.1.0.670-windows.exe
 - a. Windows may tell you that it is not safe to run this program with something like *Windows Protected Your PC*. Ignore this message. Select *More Info* and then select *Run Anyway*.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 4) When prompted to select which version of ModelSim you want to install, choose to install ModelSim Starter Edition
 - a. Leave all default options in the installer and select *Next* as necessary.



- 5) Once the installation has completed, open Quartus and navigate to *Tools->Options*. In the Options window, select *EDA Tool Options* in the left panel. In the text box next to the ModelSim label, insert the path to ModelSim's win32aloem folder. The default install location uses the path `C:\intelFPGA\19.1\modelsim_ase\win32aloem`. Click *OK* to save the simulator settings. Now you will be able to use ModelSim from the Quartus VWF editor. See the below two figures for more information.



Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

MODELSIM INSTALLATION FOR LINUX

ModelSim is a free simulator that doesn't require a license to use. We will install ModelSim 19.1, the last version of ModelSim released by Intel. This installation guide was designed for stock Ubuntu 22.04 LTS.

- 1) Go to the [ModelSim-FPGAs Standard Edition Software Version 19.1](#) and select the "Linux Software" tab of the website. (Newer versions throw errors for the `-novopt` parameter and should **NOT** be installed.)
- 2) You only need ModelSim-Intel FPGA Edition (includes Starter Edition).

ModelSim-FPGAs Standard Edition Software Version 19.1

 [Licensing Help](#)

Date- 03/31/2019

ModelSim-FPGA Standard Edition, Version 19.1std includes functional and security updates. Users should keep their software up-to-date and follow the [technical recommendations](#) to help improve security. Additional security updates are planned and will be provided as they become available. Users should promptly install the latest version upon release.

ModelSim-FPGA Standard Edition, Version 19.1std is subject to removal from the web when support for all devices in this release are available in a newer version, or all devices supported by this version are obsolete. If you would like to receive customer notifications by e-mail, please subscribe to our [subscribe to our customer notification mailing list](#).

Linux Software

Windows Software

Expand All

 [Download and Install Instructions](#)

ModelSim Software



ModelSim-FPGA Edition (Includes Starter Edition)
(ModelSimSetup-19.1.0.670-linux.run)

Size: 998.7 MB

Version: 19.1

 [Download](#)

OS: Linux*

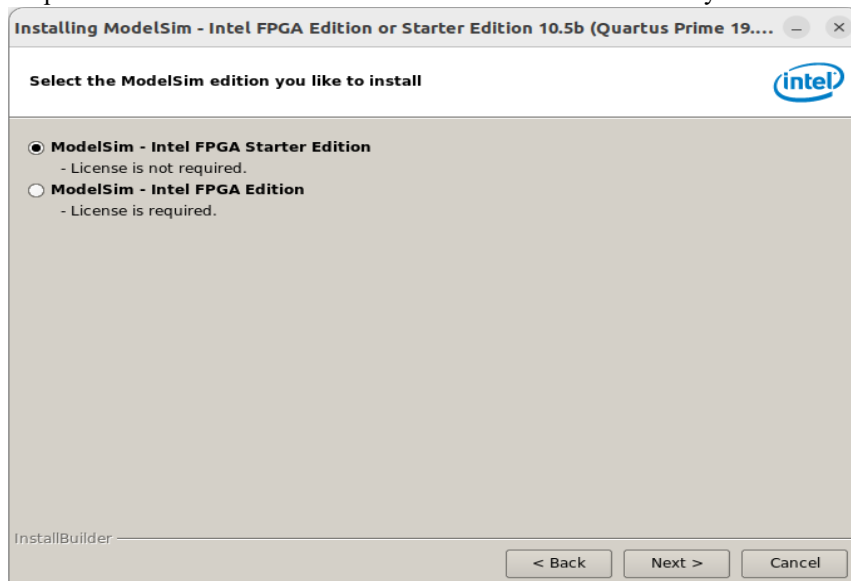
ECCN: EAR99

sha1: 354404a6fb70dd7837fe27a0cce262eada9c1d86

- 3) Open a terminal and navigate to your Downloads folder. We can then run the ModelSim installer by issuing the following commands. Make sure that you do not run the ModelSim installer as root!

```
chmod +x ModelSimSetup-19.1.0.670-linux.run  
./ModelSimSetup-19.1.0.670-linux.run
```

- 4) When prompted to select which version of ModelSim you want to install, choose to install ModelSim Starter Edition
 - a. Leave all default options in the installer and select Next each time it is necessary.

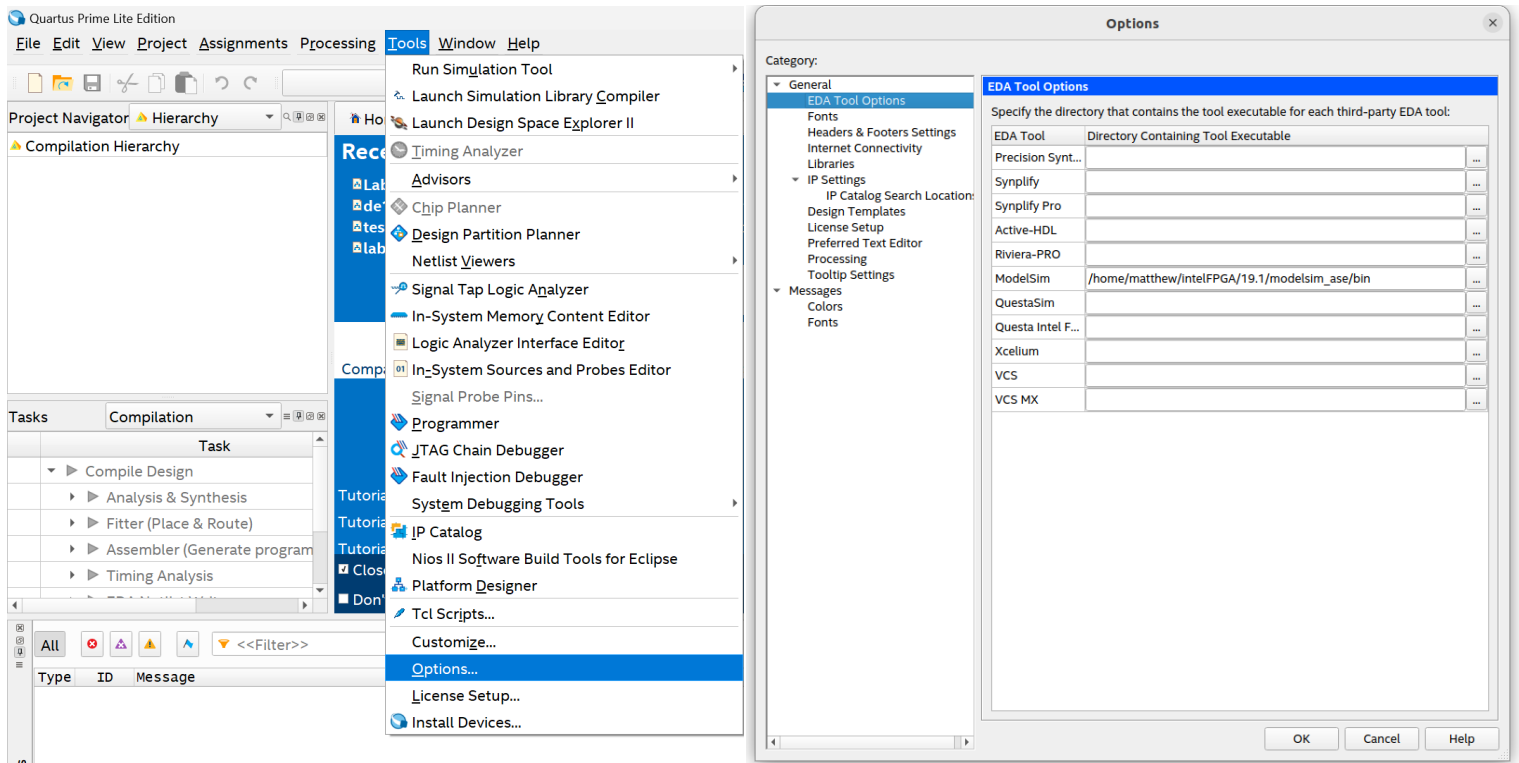


Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 5) ModelSim uses a number of old 32-bit dependencies that we must install. Luckily, these packages can still be found in Ubuntu's repositories. To install these packages, open a terminal and run:

```
sudo dpkg --add-architecture i386
sudo apt update
sudo apt install \
    libc6:i386 \
    libstdc++6:i386 \
    libncurses6:i386 \
    libx11-6:i386 \
    libxext6:i386 \
    libxft2:i386
```

- 6) To force ModelSim to use the libraries we just installed, we must edit its launch script. Use the text editor of your choice to edit `~/intelFPGA/19.1/modelsim_ase/bin/vsim` and make the following changes. You will need superuser privileges to edit this file.
- Change `mode=${MTI_VCO_MODE:-""}` to `mode=${MTI_VCO_MODE:-"32"}`
 - Change `vco="linux_rh60"` to `vco="linux"`



- 7) Once the installation has completed, open Quartus and navigate to `Tools->Options`. In the Options window, select `EDA Tool Options` in the left panel. In the text box next to the `ModelSim` label, insert the path to ModelSim's bin folder. The default install location uses the path `/home/YOURUSERNAME/intelFPGA/19.1/modelsim_ase/bin`. Click OK to save the simulator settings. Now you will be able to use ModelSim from the Quartus VWF editor.

Unfortunately, the ModelSim GUI (used in Digital Design) does not work when following these installation instructions. I'm not sure if this is due to something being broken in ModelSim or if it can be fixed by installing additional dependencies.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

APPENDICES

If you wish to install QuestaSim, the instructions are below.

NOTE: due to the Quartus Simulation Waveform Editor being outdated, Quartus injects the -novopt parameter into every simulation. On newer Simulator Versions (post 2019 from my testing), this parameter is unsupported, causing the simulation to crash.

To prevent this, you must remove the -novopt parameter for **EVERY** simulation you run. This can be found under Simulation -> Simulation Settings.

Alternatively, you can install [this shim](#) to strip this parameter from Quartus Simulations before being passed to the simulator. This effectively suppresses the issue but only works on Windows.

QUESTA INSTALLATION FOR WINDOWS 10/11

- 1) Go to the [Quartus Prime Lite 25.1 Windows download page](#) and select the “Individual Files” tab of the website. (Note that newer version may be available but have **NOT** been tested.)
- 2) You only need to download Questa - Intel FPGA Edition. You may need to accept a license agreement.

Downloads

Installer (Recommended) **Individual Files** Copyleft Licensed Source

Complete Download

Intel® Quartus® Prime Lite Edition Software (Device support included)

Download
Quartus-lite-25.1std.0.1129-windows.tar

Size: 6.5 GB
SHA1: 812e29e9d16f0359ed27d2eb5a16c42f6701cde5

** Total space required is 31.37 GB including tar file (6.51 GB), untarred files (6.51 GB) and installation (18.35 GB)
[What's Included?](#)

Individual Files

Quartus® Prime

Download
QuartusLiteSetup-25.1std.0.1129-windows.exe

Size: 1.6 GB
SHA1: 09e2c614d37d1553ae8c1e58e280b950460855da

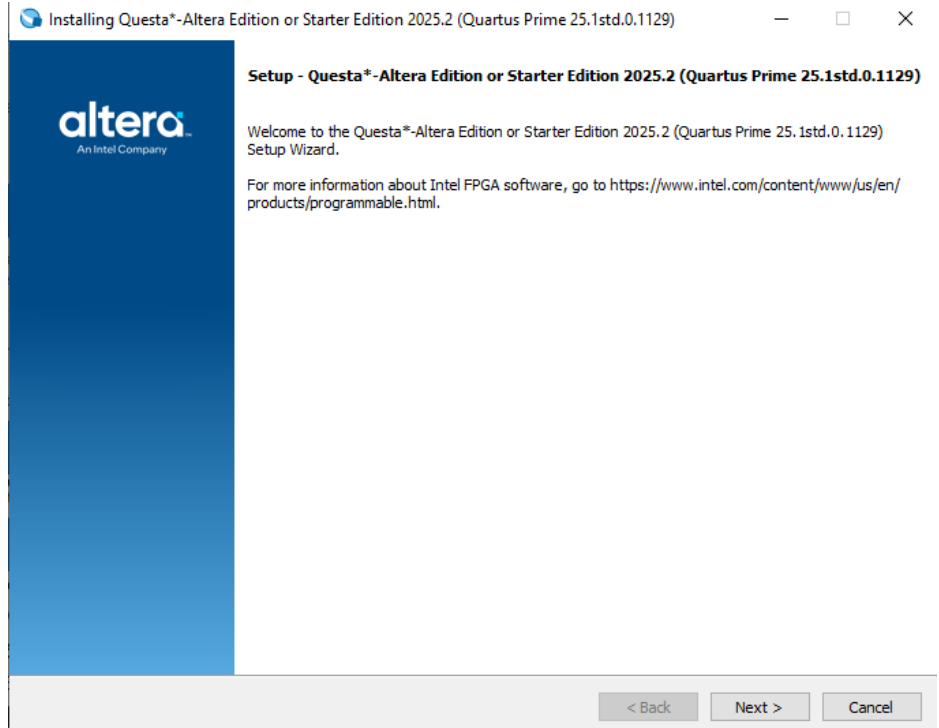
Questa®-Altera FPGA and Starter Editions

Download
QuestaSetup-25.1std.0.1129-windows.exe

Size: 942.2 MB
SHA1: 164348081f610cd652280d0e3f8a27d173b0cddf

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 3) After the Quartus software has finished downloading, run the installation file to install the Quartus software.
 - a. Windows may tell you that it is not safe to run this program with something like Windows Protected Your PC. Ignore this message. Select More Info and then select Run Anyway.
- 4) When prompted to select the version of Questa to install, choose Questa – Intel FPGA Starter Edition. Leave all default options in the installer and select Next as necessary.

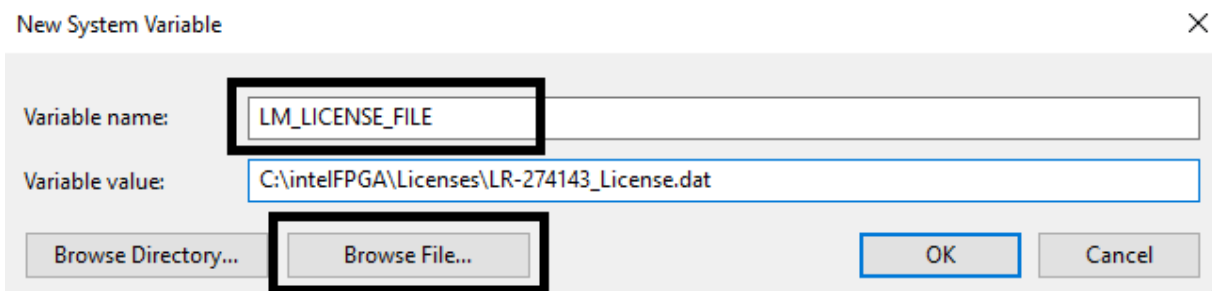
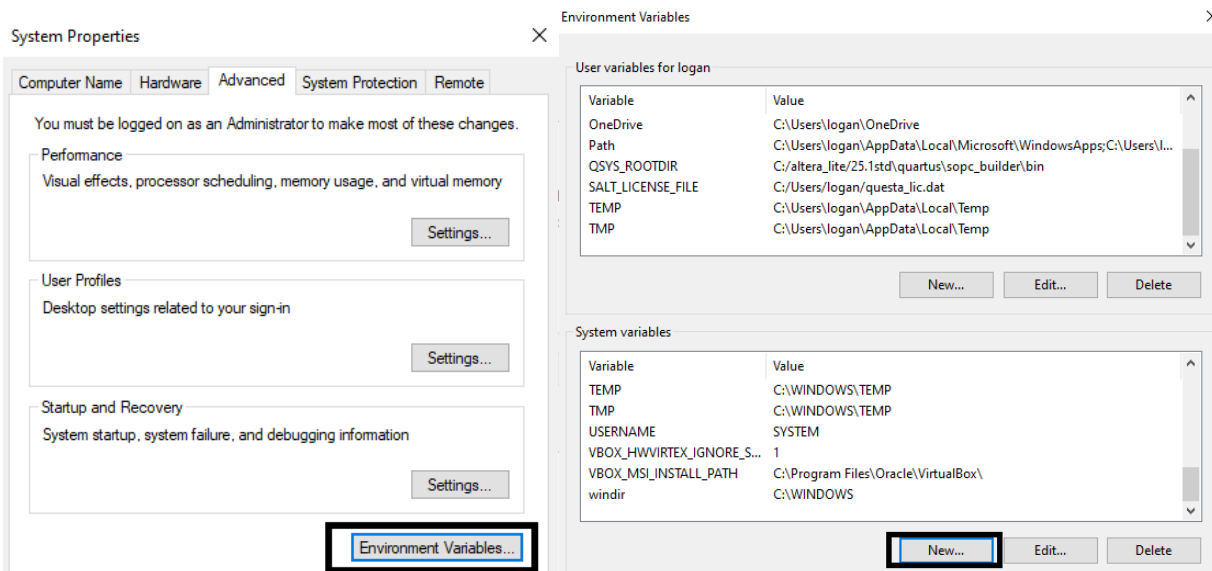


- 5) To prepare for acquiring a Questa License, we must find the hardware ID number of your network card. Intel uses this number to verify that your computer is licensed to use Questa. Open a PowerShell window and run the command `ipconfig /all`. (without the period) This will list all the network adapters connected to your computer. Look for your computer's main Wi-Fi or network card in this list and locate its Physical Address. The hardware ID number that you will put into the Intel website is your network adapter's Physical Address **with the dashes removed**. Save this number for use when requesting your Questa License. For example, the ID number in the below screenshot is `c423609f3c29`.

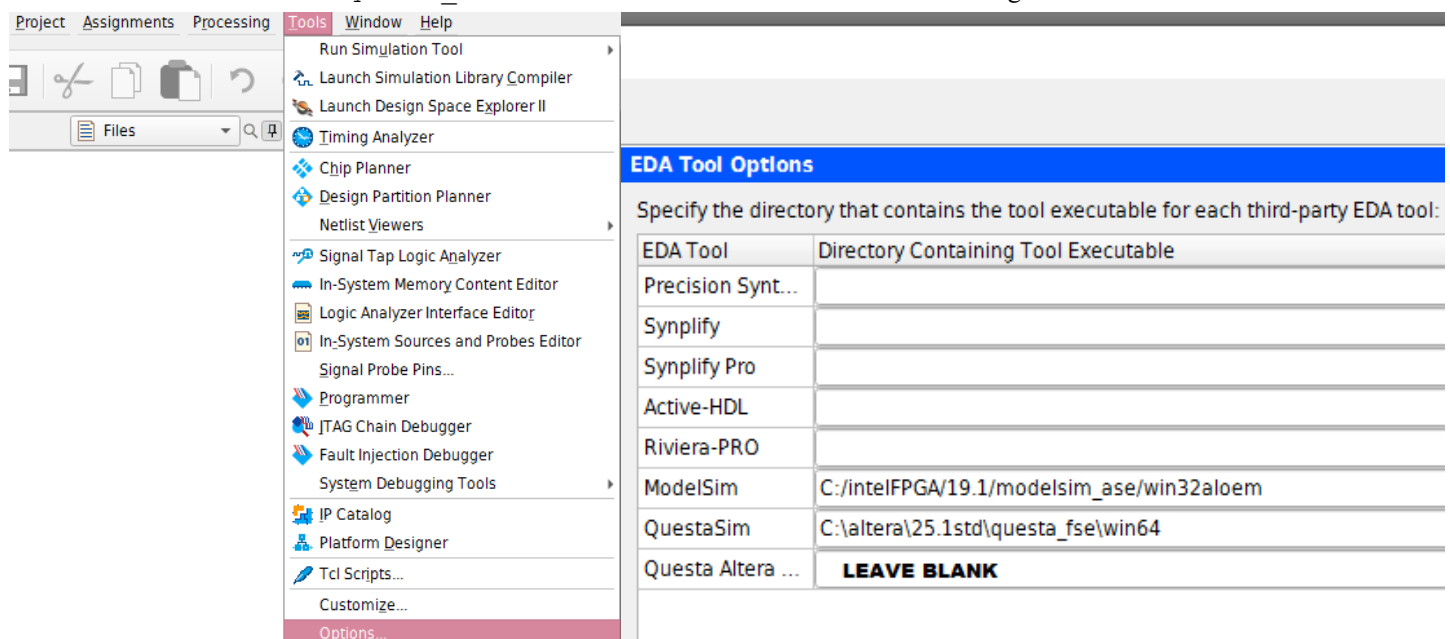
```
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : attlocal.net
Description . . . . . : Killer(R) Wi-Fi 6 AX1650s 160MHz Wireless Network Adapter
Physical Address. . . . . : C4-23-60-9F-3C-29
```

- 6) Follow the instructions in [Obtaining a Questa License](#)
- 7) Once you have obtained your Questa license, copy it to a safe folder where it won't accidentally get deleted. This tutorial assumes that you create the folder `C:\intelFPGA\licenses` and copy your license file there.
- 8) We need to set the `LM_LICENSE_FILE` (or `LM_SALT_SERVER`) environment variable globally so that Questa knows where to find the license file.
 - a. In the Start menu, search for Edit the system environment variables.
 - b. Click the Environment Variables button in the bottom right corner of the window that appears.
 - c. Under the System Variables section (bottom half of the window), click New.
 - d. In the window that appears, type in `LM_LICENSE_FILE` for the variable name. To set the variable's value, click the Browse File button in the bottom left corner of the window and navigate to where you stored your license file. Double click on the license file to save its path to the environment variable.
 - e. Click OK to confirm the new environment variable.
 - f. Restart Quartus for your changes to take effect.

Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)



- 9) Open a command prompt (type `cmd` in the Start menu) and enter `lmutil lmdiag` to confirm that your license is found. If your license cannot be verified, double check your `LM_LICENSE_FILE` (or `LM_SALT_SERVER`) environment variable.
- 10) Open Quartus and navigate to *Tools->Options*. In the Options window, select *EDA Tool Options* in the left panel. In the text box next to the Questa label, insert the path to Questa's bin folder. The default install location uses the path `C:\altera\25.1std\questa_fse\win64`. Click *OK* to save the simulator settings.



Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

QUESTA INSTALLATION FOR LINUX

The installation directions provided in this document were designed for default Ubuntu 22.04 LTS.

- 1) Go to the [Quartus Prime Lite 25.1 Linux download page](#) and select the “Individual Files” tab of the website. (Note that newer version may be available but have **NOT** been tested.)

Downloads

Installer (Recommended) Individual Files Copyleft Licensed Source

Complete Download

Intel® Quartus® Prime Lite Edition Software (Device support included)

Download Quartus-lite-25.1std.0.1129-linux.tar	Size: 8.9 GB SHA1: 833c96461eb55e0124f358c8ec9322ba1fbd5286
--	--

** Total space required is 40.10 GB including tar file (8.86 GB), untarred files (8.86 GB) and installation (22.38 GB)
[What's Included?](#)

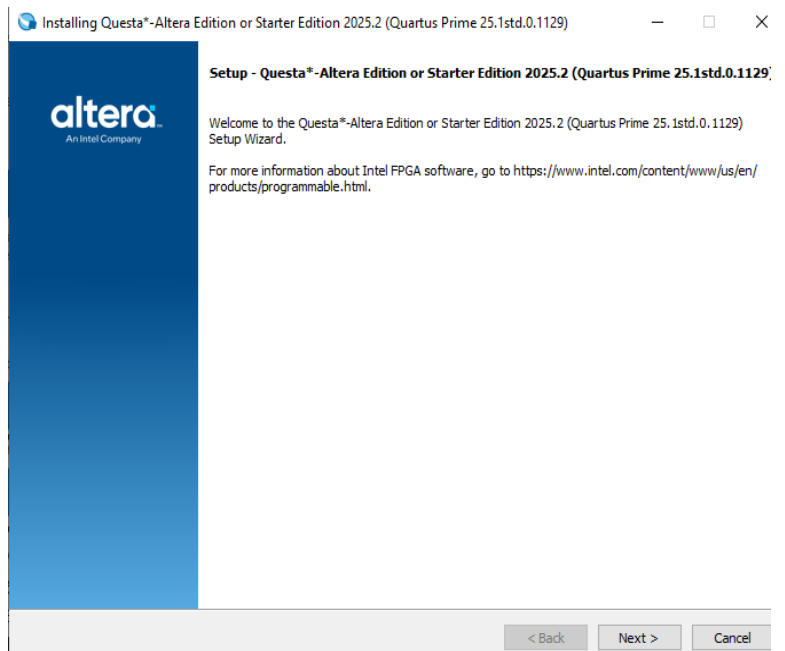
Individual Files

Download QuartusLiteSetup-25.1std.0.1129-linux.run	Size: 1.8 GB SHA1: ce0773469eacab5b7035c175484625f4ec3737d1
Download QuestaSetup-25.1std.0.1129-linux.run	Size: 3.1 GB SHA1: 149fe1e1cf253f2929804582c6cb658bca941dd5

- 2) Open a terminal and navigate to your Downloads folder. We can then run the Questa installer by issuing the following commands.
Make sure that you do not run the Questa installer as root!

```
chmod +x qinst-lite-linux-25.1std-1129.run  
./qinst-lite-linux-25.1std-1129.run
```

- 3) When prompted, choose Questa – Intel FPGA Starter Edition
 - a. Leave all default options in the installer and select Next each time it is necessary.

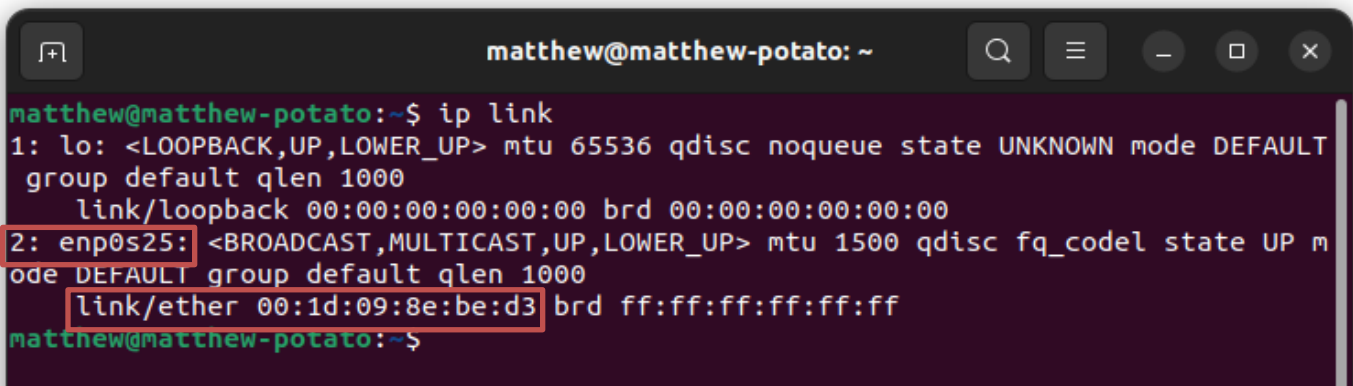


Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 4) To allow easy access to your Questa installation, create a text file called `questa.desktop` using the text editor of your choice and copy the following contents into it. Make sure to change `YOURUSERNAME` to the username of the account that you installed Questa for.

```
[Desktop Entry]
Version=1.0
Type=Application
Terminal=true
Exec=/home/YOURUSERNAME/C:/altera/25.1std/questa_fse/bin/vsim
Name=Questa
```

- If your distribution supports desktop icons, you can copy this file to your Desktop folder and it will work as a desktop shortcut
 - To add a Quartus shortcut to your Apps menu, copy this file to `~/.local/share/applications`, creating that folder if necessary. This was tested for GNOME.
- 5) To prepare for acquiring a Questa License, we must find the hardware ID number of your network card. Intel uses this number to verify that your computer is licensed to use Questa.
- Open a terminal and run the command `ip link`. (without the period)
 - Look for an adapter named `eth0` or `enp0sXX`. Under that entry, look for a line starting with `link/XXXX` followed by a series of digits and letters. This series of digits and letters (**without colons**) is your network card ID. Save this number for use when requesting your Questa License. For example, the ID number in the below screenshot is `001d098ebed3`.

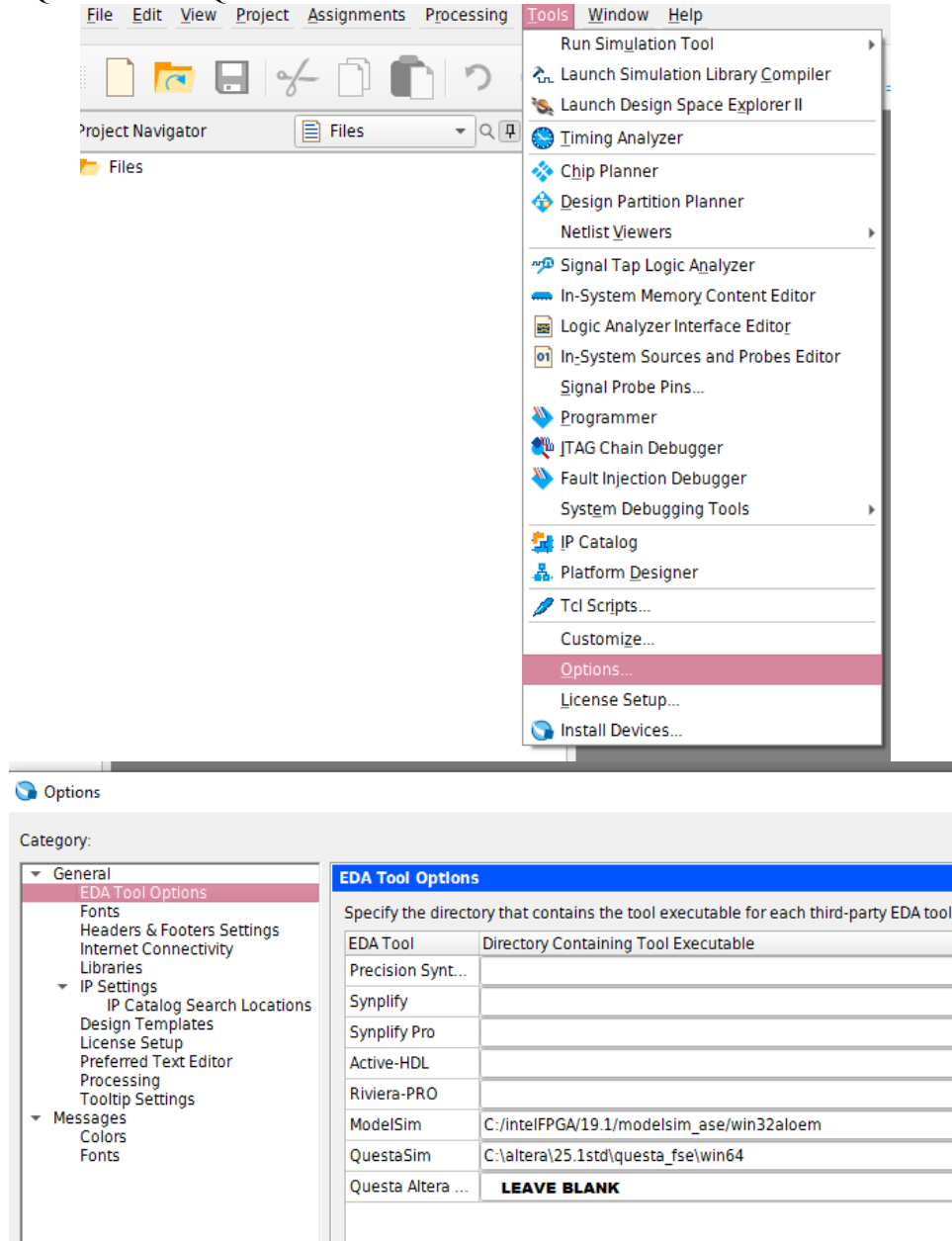


```
matthew@matthew-potato:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
   group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s25: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP m
   group default qlen 1000
    link/ether 00:1d:09:8e:be:d3 brd ff:ff:ff:ff:ff:ff
matthew@matthew-potato:~$
```

- Follow the instructions in [Obtaining a Questa License](#).
 - Once you have obtained your Questa license, copy it to a safe folder where it won't accidentally get deleted. This tutorial assumes that you create the folder `~/intelFPGA/licenses` and copy your license file there.
 - We need to set the `LM_LICENSE_FILE` environment variable globally so that Questa knows where to find the license file. To do this, create `/etc/environment.d/90quartus-license.conf` and populate it with the following text. Make sure to use the username you installed Questa for and the correct name of your license file. You will need superuser privileges to create this file.
- ```
LM_LICENSE_FILE=/home/YOURUSERNAME/intelFPGA/licenses/LR-XXXXXX_License.dat
```
- Restart your computer so that the changed environment variable takes effect.

## Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

- 10) Open Quartus and navigate to *Tools->Options*. In the Options window, select *EDA Tool Options* in the left panel. In the text boxes next to the *Questa* and *Questa – Intel* labels, insert the path to Questa's bin folder. default path: /home/**YOURUSERNAME**/intelFPGA/25.1std/questa\_fse/bin. Click *OK* to save the simulator settings. Now you will be able to use Questa from the Quartus VWF editor.



## Quartus Installation Instructions (For Quartus v25.1 and ModelSim 19.1)

### OBTAINING A QUESTA LICENSE

Questa is the newest simulator provided by Intel. It supports modern hardware and uses modern dependencies, but it unfortunately requires a free license to use. Getting a Questa license is a somewhat involved process. Only follow these instructions if you are installing Questa. If you are using ModelSim as your simulator you do not need to get a license.

- 1) Using Google Chrome or Chromium, navigate to the [Intel Self-Service Licensing Center](#). The website will not allow you to get a license if you use Firefox. Click **Enroll** for Intel® FPGA Self Service Licensing Center (SSLIC) and create an account. I would recommend signing up using your UF email. The enrollment process may require you to download the Microsoft Authenticator app onto your phone for two-factor authentication. I have not found a way to circumvent this requirement.

Intel® FPGA Self-Service Licensing Center

Home

Sign In

[Already enrolled? - Sign In here](#)

[Contact customer support](#)

[Frequently Asked Questions](#)

Enroll

If you don't already have access to the Intel® FPGA Self Service Licensing Center (SSLIC), **enroll** to get an account.

[Enroll for Intel® FPGA Self Service Licensing Center \(SSLIC\)](#)

This enrollment form is specific to external users only and not for Intel employees. Please

- 2) Once you have activated your account, go to the [Home Page for the Self-Service Licensing Center](#) and click on **Sign up for Evaluation or No-Cost Licenses**.

Intel® FPGA Self-Service Licensing Center

Home Licenses Computers and License Files Admins **Sign up for Evaluation or No-Cost Licenses** Reports Help

Hi! Welcome to the new Licensing

intel PRODUCTS SUPPORT SOLUTIONS DEVELOPERS PARTNERS  
Intel FPGA Self-Service Licensing Center  
Home Licenses Computers and License Files Admins Sign up for evaluation or No-Cost Licenses Reports Help

- 3) Click the radio button next to **Questa-Intel FPGA Starter Edition**. Set the *# of Seats* to 1.

Home Licenses Computers and License Files Admins Sign up for Evaluation or No-Cost Licenses Reports Help

Select Product & Add Additional Details Add Host & Generate License

| Web Description                                                                                                       | Maintenance Expiration | License Expiration |
|-----------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|
| <input type="radio"/> Quartus® Prime Software 90-Day Evaluation (Standard and Pro Editions) (License: EVALUATION-LIC) | 2026-03-15             | 2026-03-15         |
| <input type="radio"/> Agilent™ 3 C-Series FPGA Software Enablement (License: SW-AGILEX-3)                             | 2026-12-15             | 2026-12-15         |
| <input type="radio"/> Agilent™ 5 C-Series FPGA Software Enablement (License: SW-AGILEX-5E)                            | 2026-12-15             | 2026-12-15         |
| <input checked="" type="radio"/> Questa®-FPGA Starter Edition (License: SW-QUESTA)                                    | 2026-12-15             | 2026-12-15         |

\* # of Seats 1

Next

- 4) In the window that appears, click **New Computer**.



\* Generate License (Create a New Computer Or Choose an Existing Computer)

Choose an Existing Computer

[View All Computers](#)

Create a New Computer

+New Computer

\* ☒ I have read and agree to the terms of use of this license as listed below

Maintenance for this license is valid for 12 months from the date you sign up for this license. [Terms of Use](#)

☒ Check this box if you don't want Intel to contact you for feedback. Your feedback helps us improve the product.

- 5) Give the computer a memorable name. Under *License Type*, select **Fixed**. This creates a single-user license tied to your individual computer. Under *Computer Type*, select **NIC ID**. This will tell Intel that your computer will be identified by its network card. Under *Primary Computer ID*, type in the hardware ID of your network card (found earlier in your operating system's Questa installation

## Quartus Installation Instructions

(For Quartus v25.1 and ModelSim 19.1)

tutorial). The hardware ID should only include hexadecimal digits (0-9, a-f). **Make sure to remove any spaces, dashes, or colons in the number.** Finally, click *Save*, then *Generate*.

### Create Computer

|                           |                                                                                                          |                                                              |                                           |
|---------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------|
| * Computer Name ⓘ         | <input type="text" value="Logan's Desktop"/>                                                             | * Computer Type ⓘ                                            | <input type="text" value="NIC ID"/>       |
| * License Type ⓘ          | <input type="text" value="FIXED"/>                                                                       | ⓘ How to find your hardware information (NIC/Host/Guard ID)? |                                           |
| Companion Computer ID 1 ⓘ | <input type="text"/>                                                                                     | * Primary Computer ID ⓘ                                      | <input type="text" value="5c628bea3320"/> |
| * Primary Admin           | <div> Logan Burns</div> | Companion Computer ID 2 ⓘ                                    | <input type="text"/>                      |

- 6) Look for an email from `authorization@intel.com` sent to the address you used to create your Intel account. A license file named `LR-XXXXXX_License.dat` should be attached to the email. Download that file and return to the Questa Installation Tutorial.

## **Quartus Installation Instructions**

(For Quartus v25.1 and ModelSim 19.1)

### **Sources**

- <https://stackoverflow.com/questions/18704913/unable-to-lock-chain-insufficient-port-permissions>
- [https://wiki.archlinux.org/title/Intel\\_Quartus\\_Prime](https://wiki.archlinux.org/title/Intel_Quartus_Prime)
- [https://cdrdv2-public.intel.com/666293/quartus\\_install-683472-666293.pdf](https://cdrdv2-public.intel.com/666293/quartus_install-683472-666293.pdf)
- <https://web.archive.org/web/20220614084754/https://ecen3350.rocks/static/usb-blaster.pdf>